

Spark Homes Field Estimator

Developer Contest Submission • One-Page Writeup • July 2026

1. Most interesting design / UX decision

While mapping the official price list into groups, I found the reference implementation left 21 of its own 108 items ungrouped — things like drywall demo, haul-off, final cleaning, roof, plumbing, landscaping, and exterior paint never appeared in its checklist at all, meaning an agent using it would systematically miss real cost categories. Rather than replicate that gap, I added five extra groups (**General & Punch List**, **Plumbing**, **Grounds & Hardscape**, plus **Closet** and **Lighting** for bedroom/living rooms) so every single line item in the official price list has a home. The estimate screen is also organized by **room, not by a flat item list** — bathrooms, bedrooms, and living areas can be added or removed on the fly and are auto-numbered ("Bathroom 1", "Bathroom 2"), mirroring how an agent actually walks a house. Groups collapse by default and show a status dot so an agent can scan a whole section at a glance and know exactly what still needs review, and the running total lives in a fixed bottom bar at all times.

2. What's broken or fragile

Photos are stored as base64 inside localStorage, which is the simplest offline-safe approach but caps total capacity around 5–10MB depending on the browser — roughly 30–60 photos per device before it's worth clearing old projects. Serial-number scanning depends on Tesseract.js loading from a CDN on first use, so it only works with a connection at the moment of scanning (it fails gracefully offline, but doesn't queue for later). The regex-based serial heuristic will occasionally grab the wrong token off a busy nameplate — it's labeled a "candidate to verify," not a guaranteed read. A few of the official price list's units are inconsistent free text (e.g. "225 sqft L&M", "sqft of affected area") rather than a clean enum, so I left unit entry as free text everywhere rather than forcing it into a dropdown that couldn't represent the real data. There's also no cross-device sync: a project made on one phone doesn't appear on another, since everything is local-only by design for offline reliability.

3. Creative addition: the Deal Analyzer

A live profit-margin calculator, added as a third tab alongside Estimate and Photos. The agent enters ARV, target offer price, hold time, and monthly holding cost; the tool pulls the **repair total straight from the live checklist** (no re-entering numbers) and shows total cost basis, projected profit, and margin % with a color-coded gauge (green $\geq 20\%$, amber 10–20%, red below). I chose this over a map view or contact list because it closes the loop on the actual business question in the brief: agents aren't just cataloging repairs, they're deciding whether to make an offer. Turning the repair estimate directly into a go/no-go signal, before an agent leaves the driveway, felt like the highest-leverage thing to add.

4. What I'd ship next with two more days

(1) Move photo storage to IndexedDB and compress images on capture, to raise the practical photo ceiling well past localStorage's limits. (2) A lightweight optional sync layer (e.g. a Supabase or Firebase backend, still with local-first writes) so a team lead can see estimates roll in from multiple agents' phones in real time. (3) A comparison view across an agent's saved projects, sorted by projected margin, to help triage which properties are worth a second look. (4) Tune the serial-number OCR with a small on-device model for offline scanning instead of the CDN-loaded Tesseract.js dependency.

AI tools in my process

I used Claude throughout: to scaffold the data model (rooms/groups/items/selections), to write and iterate on the HTML/CSS/JS in a single pass given the size of the spec, and to map all 108 official price-list items into groups and cross-check that mapping against the reference app's own logic before finalizing. I made the product calls myself — the room-first layout, the two-tier pricing model (project override vs. global standard), the five added groups that close the reference app's coverage gap, and the Deal Analyzer as the creative addition — and reviewed/tested the generated code rather than shipping it unread.